

RCS CyberTrack GUI Phase 1 Implementation Report

This report documents the completion of **Phase 1: GUI Foundation & Sidebar** for the **RCS CyberTrack** security platform.

1. Files Created

We established a clean, modular React + TypeScript + Tailwind CSS structure:

- **tsconfig.json** TypeScript project settings coordinating app and node compiling references.
- **tsconfig.app.json** Compiler rules for the React app's source folder (`src`).
- **tsconfig.node.json** Compiler rules for Vite build scripts.
- **vite.config.ts** Vite configuration file loading React and Tailwind CSS v4 compiler plugins.
- **src/vite-env.d.ts** Vite client type definitions.
- **src/lib/utils.ts** Class merger utility using `clsx` and `tailwind-merge` .
- **src/app/router.tsx** Sets up `react-router-dom` for clientside dashboard and login page paths.
- **src/components/layout/AppShell.tsx** Responsive main viewport containing Sidebar, Topbar, and scrollable container.
- **src/components/layout/CyberTrackSidebar.tsx** Fully collapsible enterprise navigation sidebar with operational indicators and administrator dropdown.
- **src/components/layout/CyberTrackTopbar.tsx** Header console containing breadcrumbs, command palette search mockup, notifications, and device online heartbeat.
- **src/pages/Login.tsx** Premium dark dashboard authentication layout.
- **src/pages/Dashboard.tsx** Dashboard summary dashboard view.
- **src/pages/Firewall.tsx** Firewall rules manager tab panel.
- **src/pages/Network.tsx** IP routing and card interfaces list.

- [src/pages/Devices.tsx](#) Active network device scanning inventory.
 - [src/pages/Audit.tsx](#) System audit logs database table.
 - [src/pages/Settings.tsx](#) Appliance preferences control panel.
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2. Files Modified

- [package.json](#) Added TypeScript, Routing, and Tailwind CSS v4 dependencies. Adjusted build commands.
 - [index.html](#) Updated script loader source to reference `/src/main.tsx`.
 - [src/index.css](#) Injected Google Fonts (Inter & JetBrains Mono), imported Tailwind CSS v4, and defined the global theme tokens.
 - [src/main.tsx](#) Mounted `RouterProvider` to the react root rendering element.
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3. npm Packages Installed

- `typescript` (and `@types/react`, `@types/react-dom`, `@types/node`): For compiler support and type safety.
 - `tailwindcss` & `@tailwindcss/vite`: For compiling Tailwind CSS v4 with native Vite build speeds.
 - `react-router-dom`: For routing.
 - `lucide-react`: For unified, clean vector security/network interface icons.
 - `clsx` & `tailwind-merge`: For conditional style class utilities.
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4. Commands to Run

Inside the `/media/kamran-shekh/WorkData/Projects/Rahul/rcs cybertrack/gui` folder:

- **Start Dev Server:**

```
npm run dev
```

- **Verify TypeScript and Build Assets:**

```
npm run build
```

- **Lint Check:**

```
npm run lint
```

5. Exact Localhost URL

- Local Host URL: `http://localhost:5174/` (Vite starts on port `5174` because port `5173` is occupied).

6. Architecture & Theme Explanation

Theme System

- Built custom CSS vars matching the design tokens inside the `@theme` directive in Tailwind CSS v4.
- Applied dark-first backgrounds (`#070B14` , `#0A0F1C` , `#0F172A`) and custom typography rules (`Inter` for standard labels, `JetBrains Mono` for system values like logs, rules, and IP addresses).

Component Layout

```
+-----+
|               CyberTrackTopbar               |
| [Menu] [Breadcrumbs]      [Search] [Appliance] [Bell] [User]|
+-----+
```

CyberTrackSidebar		
[Shield Logo]	Main Viewport Area	
RCS CyberTrack		
	(renders child route	
- Dashboard	pages, e.g. Audit Logs,	
- Firewall	Firewall Rules, etc.)	
- Network		
- Devices		
- Audit Logs		
- Settings		
[Core: Online]		
[Admin details]		
Version v0.1.0		
+-----+	+-----+	+-----+

- **State Management:** Collapsible/expandable state is hoisted to `AppShell.tsx` and propagated down as props. On mobile viewports, responsive triggers slide open a mobile drawer layout dynamically.
- **Router Routing:** `app/router.tsx` handles dynamic client paths, mapping components to their respective dashboard slots.